

Review: container-derivative uniqueness and tangent-structure notes (MacBeth 5f1531d)

Rick

To: MacBeth (Kodamai)

2026-10-07

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Reviewed: *A uniqueness theorem for the container derivative* (13 pp., “revised 5 Oct”)
The container derivative is a tangent structure (9 pp., dated 3 Oct)
both as attached to your email of 2026-10-06, said to be at your commit 5f1531d
WIP commit: 7e8dee8
Check script: `reviews/scripts/2026-10-07-macbeth-checks.py` (sympy, <2 s)

Labels: ERROR (claim false or proof broken), GAP (missing step that can plausibly be filled), WORDING, MINOR. Page numbers are the printed page numbers. “U” is the uniqueness note and “E” is the existence (tangent-structure) note.

Summary

The mathematical core of U is now right, in the category you now name. I re-derived the counts in finite free $\mathbb{N}$ -modules, with $T = (-) \otimes_{\mathbb{N}} D$, $D = \mathbb{N} \oplus M$, $\text{rank } M = n$: $\text{rank } D_2 = 1 + 2n$ and $\text{rank } V(\mathbb{N}) = 1 + n + n^2$. These agree only for $n \in \{0, 1\}$; see the table in Appendix A. So the *necessity* half of Theorem 1/18 holds there, modulo small gaps.

Three things still overstate.

- (a) The bridge from $\mathbb{N}$ -modules back to $\triangleleft$ and “the container derivative” (Lemma 5 and everything that leans on it) is still not valid at the level of *morphisms*.
- (b) The existence half is cited to E, but E contains no $(-) \otimes W$ statement. E was not revised: its PDF CreationDate is 2026-10-03, and your commit message for 5f1531d says “existence note: already honest ... no change needed”.
- (c) U’s definition of “solid” contradicts its own Definition 2. As literally written, $M = y$ is *not* solid.

I also think F4 (the $\mathbb{N} \cdot y$ question) can probably be *closed negatively* by the canonical-flip axiom (Finding U7). Please check this rather than take my word for it.

Verdicts

Item	Verdict	Reason (finding)
U Lemma 5 (three tensors)	proved on objects only	true for shape counts; false as an identification of operations with their morphisms (U1)
U Lemma 7 (rank invariant)	proved	
U Lemma 8 / Thm 9 (finite rigidity)	false as stated; proved after a fix	$\mu _{M \otimes M} = 0$ by Def. 2, so y is not “solid”; fix: define solidity via ℓ (U3)
U Prop 10 ($\mathbb{N} \cdot y$ solid)	not proved in a useful sense	abstract iso only; fails flip compatibility; outside “finite-rank” (U7)
U Lemma 12 ($\triangleleft$ preserves pullbacks)	proved, but misapplied	true in Poly; §6 needs the $\mathbb{N}$ -module statement (U5)
U Lemma 13 (three ranks)	proved in $\mathbb{N}$ -Mod	not in Poly’s linear layer, where $ P' = 1 + n$ (U1)
U Thm 14 (converse)	proved-modulo-gap	$p, 0$ must be induced by D ; $T_2 = (-) \otimes D_2$ justification; conclusion “solid” (U3, U5)
U Thm 1/18, $\mathbb{N}$ -module sentence	proved-modulo-gap	necessity OK; existence for $D = W$ not in [11] (U2); “exactly two <i>structures</i> ” needs uniqueness of $+, \ell, c$ (U6)
U Thm 1/18, “Equivalently ... container derivative” sentence	not proved as phrased	wrong bridge; and on linear maps ∂ is trivial (U1, U4)
E Thm 1/4 (SPoly is a CDC with ∂)	proved-modulo-gap	only after “finitary” is changed to “finite-support” (E1)
E Prop 9 (linear maps = $S \cdot y$)	proved	checked
E Thm 11 (restriction to directed containers)	not proved	sketch: Poly/ D undefined, 2-cells missing (E2)

1 Scope fix: is it implemented everywhere?

No. I grepped the extracted text of both PDFs for the old scope. The old identification “ $(-) \triangleleft D$ is the dual-number substitution” is gone from U. What remains are these leftover or contradictory statements:

- S1.** ERROR/WORDING U title page, subtitle: “ ∂ is the only nontrivial representable tangent structure on *finite-support polynomial functors*”. This is the old scope verbatim. The theorem is about finite free $\mathbb{N}$ -modules.
- S2.** ERROR U p.5, Def. 2: “on a polynomial p , base change by W is the dual-number substitution $T(p) = p(y + \varepsilon v) \bmod \varepsilon^2$ ”. Repeated on p.5 (1.238) and in the abstract. This is false. Base change $\mathbb{N}[y] \rightarrow W[y]$ sends $y^2 \mapsto y^2$. The element $y^2 + 2\varepsilon v y$ is the image of y^2 under the *substitution* $y \mapsto y + \varepsilon v$, which is a map $\mathbb{N}[y] \rightarrow \mathbb{N}[y, v][\varepsilon]/\varepsilon^2$. That is the unit of the adjunction between $(-) \otimes W$ and the Kähler functor $A \mapsto \text{Sym}_A \Omega_A$, not base change of p (script item 1). There is also a type mismatch: in the category of U, the objects are modules, and a non-linear polynomial p is neither an object nor a morphism there.
- S3.** ERROR U p.5, 1.241: “That $T = (-) \otimes W$ is a genuine tangent structure ... is the content of [11]”. Also p.3 (“representability [11]”) and §7 (“the full ... verification is the content of [11]”). E never mentions $\otimes W$, Weil algebras as tangent functor, or $\mathbb{N}$ -modules. E’s tangent category is SPoly, with objects arities n and $T(n) = 2n$. See U2 for how to repair this.
- S4.** WORDING U p.10, Rem. 15: “In Poly that fibre is a direct coproduct”. U p.8, Rem. 11: “whether uniqueness ... fails over all of Poly is left open”. Both are old-scope phrasing.

- S5.** MINOR U abstract and Rem. 4 say that on all of Poly “only $T = \text{id}$ survives”. That is shown only for *finite* rank, where $(1 + n)^2 = 1 + n$. In honest Poly, infinite κ passes the rank test ($\kappa^2 = \kappa$). Say “only id among finite-rank D ”.
- S6.** WORDING Email: “Attached are the rewritten uniqueness note and its companion existence note, incorporating the scope fix”. E is unchanged since 3 Oct. Your own commit message says so, so this is a description slip, not a hidden change. Still, Strathclyde listeners will read E as the existence half of U’s theorem, and it is not (S3).
- S7.** ERROR U p.6, “Why the infinitesimal part is linear”: “So linearity of M is forced, not assumed”. This is the same heuristic my F5 objected to (“no vector-space fibre”; over $\mathbb{N}$ there are no vector spaces). Meanwhile Def. 2 *assumes* M finite free, and pp.4, 10 list “that M is linear” as a cited framework fact. Of the three, keep the hypothesis. In the $\mathbb{N}$ -module setting linearity is automatic, so delete the paragraph. F5 is therefore *not* incorporated, despite the email.

2 Uniqueness note: findings

- U1.** ERROR **Lemma 5 identifies objects, not categories.** (p.6; used in the abstract, p.2 Thm 1 “Equivalently”, p.4, §6 preamble p.8, §9.) $(S \cdot y) \triangleleft (T \cdot y)$ and $\mathbb{N}^S \otimes \mathbb{N}^T$ both have $|S \times T|$ generators, so the statement is true on objects. But the two hom-sets differ:

- A Poly morphism $S \cdot y \rightarrow T \cdot y$ is a *function* $S \rightarrow T$ (your Lemma 7 proof says so).
- An $\mathbb{N}$ -module map $\mathbb{N}^S \rightarrow \mathbb{N}^T$ is an $\mathbb{N}$ -matrix.

The augmentation $W \rightarrow \mathbb{N}$, $\varepsilon \mapsto 0$, sends a basis element to 0. It is not a function on shapes, so it is not a Poly morphism. Yet it is exactly the map whose fibre gives $|P'| = 1 + n + n^2$. With Poly morphisms, $|P'| = 1 + n$ (Appendix A, as in my 10-05 report). So “we compute with the quick substitution formula $\triangleleft \dots$ legitimate precisely because every object in sight is linear” (p.8) is not legitimate: the *morphisms* in sight are not Poly morphisms.

Repair (I think this is the bridge you want). Use E’s SPoly, not Poly. The subcategory $\text{Lin}(\text{SPoly})$ of *linear maps* $n \rightarrow m$ consists of m -tuples $\sum_j S_{ij} \cdot y_j$, i.e. $\mathbb{N}$ -matrices, with substitution as matrix product. It is equivalent to finite free $\mathbb{N}$ -modules. The CDC tangent structure of SPoly restricts to it: $T(f) = f \times f$ for linear f , and $p, 0, +, \ell, c$ are all linear. Under this equivalence it is exactly $(-) \otimes W$ with $p = (-) \otimes \text{aug}$. So the correct sentence is: “on the subcategory of linear maps of SPoly, the CDC tangent structure is $(-) \otimes W$, and it is one of exactly two of this form”. Here $S \cdot y$ appears as a *morphism* (1×1 matrix), not as an object, and $\triangleleft$ appears as composition, not as a tensor of objects.

- U2.** GAP **Existence in the stated category.** (§7, p.11.) As cited, existence rests on [11], which proves something else (S3). It is easy to fill in either of two ways:

- Use the restriction argument in U1. Pullbacks T_n and V in Lin are the same products as in SPoly, and the induced maps from linear cones are linear tuples.
- Or note directly that on any category of $\mathbb{N}$ -modules $A \mapsto A \oplus A$ is the biproduct tangent structure of a semiadditive category, with $D[f] = f\pi_2$. For W , the canonical $v: T_2 \rightarrow V$ is $(a, b, c) \mapsto a + \varepsilon_{\text{in}}c + \varepsilon_1\varepsilon_2b$, which is bijective.

Either way, cite Cockett–Cruttwell for the semiadditive/CDC example, not [11].

U3. ERROR Definition 3 makes W non-solid. (p. 5, Def. 2 vs Def. 3; abstract; Thm 9 p. 7; Thm 14 conclusion p. 10.)

- Def. 2 requires $M \cdot M = 0$ in D , so the multiplication $\mu: M \otimes M \rightarrow M$ is the zero map.
- Def. 3 calls M solid iff μ is an isomorphism, which holds only for $M = 0$.
- For W : $\mu(\varepsilon \otimes \varepsilon) = \varepsilon^2 = 0$.

So “ $M = y$ is solid” (Thm 9) and “ μ is an isomorphism in each case” (Thm 14, last line) are false as written. The map that *is* an isomorphism for W is the vertical-lift component $\ell_M: M \rightarrow M_1 \otimes M_2$, $\varepsilon \mapsto \varepsilon_1 \varepsilon_2$. Universality forces exactly this (see U7). I suspect this is what Lanfranchi–Lemay’s solidity encodes, but I have not re-read their definition, so please check it. With solidity redefined as “ ℓ_M is an iso”, Lemma 8 and Thm 9 go through unchanged, because only the abstract iso $M \otimes M \cong M$ is used.

U4. WORDING, IMPORTANT FOR THE AUDIENCE On the linear layer ∂ is trivial. (Thm 1 p. 2, §1 “Contribution” p. 4, §8, §9.)

- On finite free $\mathbb{N}$ -modules, $(-) \otimes W$ is $A \mapsto A \oplus A$ and $T(f) = f \oplus f$.
- The induced combinator is $D[f] = f \circ \pi_2$. That is your own E Prop. 9: linear maps are exactly those with $\partial[f] = f \pi_2$.
- None of the one-hole-context content of ∂ (the $y^n \mapsto ny^{n-1}$ part) is visible in the category where the classification takes place.

“The container derivative is the only nontrivial representable first-order tangent structure” therefore needs the qualifier “... whose restriction to linear maps is ...”, or better, a plain statement: “the tangent structures $(-) \otimes D$ on finite free $\mathbb{N}$ -modules are exactly $D = \mathbb{N}$ and $D = W$; the second is the restriction of the SPoly/∂ structure”. The Strathclyde audience includes the AAGM community, so expect this question first. Likewise, “answers Cockett’s question in its strongest form: ... differentiation is the only nontrivial way” (p. 13) does not follow. Nothing here classifies tangent structures on SPoly itself.

U5. GAP Theorem 14: hidden hypotheses and the wrong lemma. (pp. 8–10.)

- (i) The proof uses that $p = (-) \otimes \text{aug}$ and $0 = (-) \otimes \text{unit}$, i.e. that the bundle data come from D . Def. 3 only says that the *functor* is $(-) \otimes D$. A natural map $(-) \otimes D \Rightarrow \text{id}$ is any module map $D \rightarrow \mathbb{N}$. Make “representable” include the induced $p, 0$ (and $+, \ell, c$), as in Lanfranchi–Lemay.
- (ii) $T_2 = (-) \otimes D_2$ and $V = (-) \otimes P'$ are justified by Lemma 12, which is about pullbacks in Poly . In $\mathbb{N}\text{-Mod}$ you need instead that $A \otimes -$ preserves these particular pullbacks. For A free this is immediate, since the pullbacks are split coordinate pullbacks.
- (iii) The computation of P' as a preimage needs that, in a free $\mathbb{N}$ -module, an element lies in the summand $A \otimes \mathbb{N}$ iff its other coordinates vanish. That is true, and it is worth one sentence: it is the “no subtraction” point, done properly.

Granting (i), the argument is correct. Evaluating at $\mathbb{N}$, ranks are complete invariants, and cancellation gives $n^2 = n$.

U6. GAP/MINOR “Exactly two” counts objects D , not structures. (Thm 18, p. 11.) You should also show that, for $D \in \{\mathbb{N}, W\}$, the remaining data $+, \ell, c$ are forced. Over $\mathbb{N}$ this is quick:

- The unit laws force $+$ to be the fold on $M_1 \oplus M_2$.

- ℓ_M must be an iso $\mathbb{N} \rightarrow \mathbb{N}$, hence the identity.
- c on $M_1 \otimes M_2$ is an involution of $\mathbb{N}$, hence the identity.

Add the paragraph.

- U7.** MINOR Rem. 16 still says “solidity coincides with indecomposability”. That was my earlier minor point and it is still unsupported. Also, on p. 3 Lanfranchi–Lemay work on the opposite of commutative *algebras* (affine schemes), not “the opposite of the category of modules”. On that side the tangent functor is the exponential $(-)^{\text{Spec } D}$; your $(-) \otimes D$ is its left-adjoint/algebra-side avatar (Cockett–Cruttwell’s adjoint tangent structure). Please say this, because a tangent-category referee will ask.
- U8.** SUGGESTION; PLEASE CHECK **F4 looks closable: the canonical flip excludes $\mathbb{N} \cdot y$** . (§5, Rem. 11, Rem. 17.) F4 is honestly marked open in all five places: abstract, p. 3, Rem. 11, Table 1, §9. That part is fine. But first note that $\mathbb{N} \cdot y$ is not a finite-rank module, so F4 lies outside U’s own category. It has to be posed on all free $\mathbb{N}$ -modules. There, the following argument should settle it:

- (i) The axioms that ℓ is a bundle map and that $p_T \ell = 0p$ force $\ell = 1 \oplus \ell_M$ with $\ell_M: M \rightarrow M_1 \otimes M_2$.
- (ii) The canonical $v: T_2 \rightarrow V$ is then $(a, b, c) \mapsto a + \varepsilon_{\text{in}} c + \ell_M(b)$. So universality holds iff ℓ_M is an isomorphism. This is the correct “solidity”, and it is stronger than an abstract rank equality.
- (iii) For a representable structure c is the swap σ of $D \otimes D$, and Cockett–Cruttwell require $c\ell = \ell$, i.e. $\sigma\ell_M = \ell_M$.
- (iv) Isomorphisms of free $\mathbb{N}$ -modules permute bases (the atoms). So $\ell_M(e_k) = e_i \otimes e_j$ for some i, j , and $\sigma\ell_M = \ell_M$ forces $i = j$.
- (v) Surjectivity then forces every $e_i \otimes e_j$ to be diagonal, so $\text{rank } M \leq 1$, *for every cardinality*.

Script item 4 confirms there is no flip-compatible iso for ranks 2, 3, 4. The Cantor-pairing ℓ_M for $\mathbb{N} \cdot y$ is not symmetric, so $(-) \otimes (\mathbb{N} \oplus \mathbb{N}^{(\mathbb{N})})$ is *not* a representable tangent structure. If this survives your check:

- Uniqueness holds for free M of any rank.
- Prop. 10’s “finiteness is sharp” becomes an artefact of measuring solidity by an abstract iso, and should be withdrawn.
- $\mathbb{N}$ has nothing to do with it: the argument is the familiar fact that solid objects have symmetric tensor square, which is consistent with Lanfranchi–Lemay’s $\mathbb{Z} \ltimes \mathbb{Q}$, where $\mathbb{Q} \otimes \mathbb{Q} \cong \mathbb{Q}$ is symmetric.

My caveat: this assumes c is the swap, which is the representable convention. A non-representable choice of c on $M_1 \otimes M_2$ would have to be the identity there, and I have not checked the remaining axioms for that variant. This also gives a second, one-line proof of the finite case, so the rank-count converse is not needed for uniqueness. It remains a nice “no-subtraction” remark.

3 Existence note: findings

- E1.** GAP (DEFINITION) “**Finitary**” vs “**finite-support**”. (p. 3, Def. 2; p. 2, “The result”.) SPoly is defined with *finitary* containers, meaning finite position sets but possibly infinitely many shapes. Then $N(F) \notin \mathbb{N}[m_1, \dots]$: for $F = \mathbb{N} \cdot y$, $N(F) = \aleph_0 \cdot m$. So Def. 3 and Lemma 6 are undefined

on part of SPoly. U (p. 4) defines SPoly as *finite-support*, which is inconsistent with E. Restrict E to finite support, i.e. finitely many shapes, each with finitely many positions. Then Lemmas 5–7 are fine. I checked:

- faithfulness, because a polynomial function on $\mathbb{N}^n$ determines its coefficients;
- the CD.6 and CD.7 forms as you state them, symbolically on test polynomials (script item 7).
Your CD.6 with b free is valid, since the second-order term carries the factor $x' = 0$.

The tangent axioms (additive bundle, lift, flip, universality) are all *inherited* from Cockett–Cruttwell’s $\text{CDC} \Rightarrow \text{tangent}$ result. None is checked directly, which is legitimate. Please verify the proposition number “[5, Prop. 4.7]” (I did not).

E2. ERROR (LABELLING) Theorem 11 is a sketch, not a theorem. (pp. 7–8.)

- “ $(\text{Poly}/D, \triangleleft_D, y + \varepsilon v)$ ” is never defined.
- SPoly is built on isomorphism classes, so T has no action on the container *morphisms* (2-cells). But comonoid structure maps are 2-cells.
- Comonoid preservation is “borrowed” from an unspecified 2-functoriality in [6].
- Remark 12 itself shows the Set reading fails.

What is actually proved is that T is functorial for substitution: $T(c \circ d) = T(c) \circ T(d)$, which is CD.5. Relabel it as a Proposition-sketch or Conjecture, and move “the tangent bundle of a category lives over the dual numbers” (abstract, §6, §7) out of the headline claims.

E3. MINOR E is dated 3 Oct and carries no revision note, so its relation to U’s corrected scope is invisible to a reader. Add a sentence: “E’s tangent category is SPoly ($T(n) = 2n$); its restriction to linear maps is U’s $(-) \otimes W$ (see U1)”. Also, “every object is linear, so the question is empty” (p. 6) is a bit glib: in a CDC $T(A) = A \times A$ by construction, so the question is not asked, rather than answered.

4 What you can safely say at Strathclyde

1. Finite-support polynomial functors (SPoly, objects = arities) form a Cartesian differential category with combinator the AAGM derivative ∂ . The proof is by reflection along the faithful counting functor into $\mathbb{N}$ -polynomials, so SPoly is a tangent category with $T(n) = 2n$. Honest novelty: near-folklore (“polynomials over a rig”), but a clean statement and a clean proof.
2. The linear maps of SPoly are exactly the linear containers $S \cdot y$ (Prop. 9). This is the non-vacuity check.
3. On finite free $\mathbb{N}$ -modules (equivalently, linear maps of SPoly), the tangent structures of the form $(-) \otimes D$, with D augmented square-zero and $p, 0$ induced, are exactly $D = \mathbb{N}$ (trivial) and $D = W$ (the restriction of the ∂ structure). The proof is a rank count over a rig with no PID and no subtraction. Say “restriction of the ∂ structure”, not “ ∂ is the unique tangent structure on polynomial functors”.
4. In full $(\text{Poly}, \triangleleft)$, right substitution $(-) \triangleleft D$ never gives $\partial(y^2 \triangleleft 2y = 4y^2)$. Among finite-rank D only the identity survives.
5. Open: tangent structures on SPoly itself, which are not classified; the reverse/lens reading; and Problem 13.

Do not say:

- “ ∂ is the only nontrivial tangent structure on finite-support polynomial functors” (U subtitle).
- “base change of p by W is $p(y + \varepsilon v)$ ”.
- “ $\otimes, \triangleleft$ and Dirichlet coincide on the linear layer” as a categorical statement.
- “finiteness is sharp” or “ $\mathbb{N} \cdot y$ is solid” (see U7).
- “ T restricts to directed containers” as a theorem.
- “Cockett’s question answered in its strongest form”.

A Computations (script output)

Finite free $\mathbb{N}$ -modules, $D = \mathbb{N} \oplus M$, $\text{rank } M = n$, $T(p)$ applied to the inner factor. The Poly-function model has linear containers with morphisms equal to functions on shapes.

n	$\text{rk } D \otimes D$	$\text{rk } D_2$	$\text{rk } V$	iso?	Poly $ D \times_y D $	Poly $ P' $
0	1	1	1	yes	1	1
1	4	3	3	yes	4	2
2	9	5	7	no	9	3
3	16	7	13	no	16	4
4	25	9	21	no	25	5
5	36	11	31	no	36	6

Other checks:

- $y^2 \mapsto y^2 + 2\varepsilon v y$ under substitution, versus $y^2 \mapsto y^2$ under base change $\mathbb{N}[y] \rightarrow W[y]$; the same holds for y^3 and $y^2 + 1$.
- Remark 6: $(y + 1) \triangleleft (y + 1) = y + 2$ and Dirichlet $(y + 1) \otimes (y + 1) = y + 3$ (correct).
- E Prop. 9: the constraints are $a_0 = a_2 = a_3 = a_4 = 0$ (correct).
- A flip-compatible basis bijection $[n] \rightarrow [n]^2$ exists only for $n \leq 1$.
- U’s Lean arithmetic identities $((1+n)^2 = \text{rk } P' + n = \text{rk } D_2 + n^2)$ are correct, but they are not the load-bearing point. All of the issues above are about which category the numbers live in, not about the arithmetic.